$$\boxed{y(n) = x(n) * h(n)} \quad ; \quad h(n) = \delta(n) - \delta(n-4)$$

☆ La convolución con δ tiene la propiedad:

$$x(n) * \delta(n-k) = x(n-k)$$

$$y(n) = x(n) * \delta(n) - x(n) * \delta(n-4) = x(n) - x(n-4)$$

$$\boxed{y(n) = x(n) - x(n-4)}$$

<u>Prop</u>: $\cos\alpha - \cos\beta = -2\,\text{sen}\left(\frac{\alpha+\beta}{2}\right)\text{sen}\left(\frac{\alpha-\beta}{2}\right)$

ⓐ $x(n) = \cos(\omega_0 n \cdot T_s)$

$$y(n) = \cos(\underbrace{\omega_0 n \cdot T_s}_{\alpha}) - x(\underbrace{\omega_0 (n-4) \cdot T_s}_{\beta})$$

$$\frac{\alpha+\beta}{2} = \frac{\omega_0 n T_s + \omega_0 (n-4) T_s}{2} = \omega_0 (n-2) T_s \quad ; \quad \frac{\alpha-\beta}{2} = 2\omega_0 T_s$$

$$y(n) = -2\,\text{sen}(\omega_0 (n-2) T_s)\,\text{sen}(2\omega_0 T_s) \longrightarrow \underline{\text{prop}}: \; -\text{sen}\,\theta = \cos\left(\theta + \frac{\pi}{2}\right)$$

$$y(n) = 2\,\text{sen}(2\omega_0 T_s)\cos\left(\omega_0 (n-2) T_s + \frac{\pi}{2}\right)$$

$$\boxed{y(n) = \underbrace{2\,\text{sen}(2\omega_0 T_s)}_{A}\cos\left(\omega_0 n T_s \underbrace{- 2\omega_0 T_s + \frac{\pi}{2}}_{\phi}\right)}$$

ⓑ $x(n) = \left(\frac{1}{2}\right)^n u(n)$

$$y(n) = \frac{1}{2^n} u(n) - \left(\frac{1}{2}\right)^{n-4} u(n-4) = \boxed{\frac{1}{2^n}\left(u(n) - 16\,u(n-4)\right)}$$

ⓒ $\boxed{x(n) = u(n+1) - u(n-2)}$